Pranshav Patel

github.com/pranshavpatel | linkedin.com/in/pranshavpatel | pranshavpatel1803@gmail.com | (646) 337-8592 | Raleigh, NC

Education

North Carolina State University Aug 2025 – May 2027
Master of Computer Science *GPA: 4.0/4.0*
• Coursework: Design and Analysis of Algorithms, Software Engineering, Automated Learning and Data Analysis, Advanced Topics in Machine Learning and Neural Networks, Computer Networks, DevOps

Nirma University Sep 2021 – Jun 2025
Bachelor of Technology in Computer Science *GPA: 8.58/10.0*
• Coursework: Machine Learning, Deep Learning, Natural Language Processing, Database Management Systems, Probability and Statistics, Data Structures and Algorithms, Software Engineering

Work Experience

Data Science Intern, *LexisNexis Legal & Professional* *Jun 2026 – Aug 2026*
- Re-architected an **agentic LLM pipeline** for legal due diligence (**Python, Google ADK, LiteLLM, FastAPI, Pydantic**), replacing a nondeterministic LLM planning step with a deterministic, **budget-capped evidence-retrieval layer** driven by versioned JSON review plans.
- Cut **LLM prompt tokens by 66%, inference cost by 63%, and end-to-end latency by 38%** (mean across 20 controlled A/B experiments) while keeping output quality on par with the production baseline.
- Designed a reproducible **A/B evaluation harness** with frozen inputs, isolated per-arm processes, input hashing, automated quality gates, and **LLM-as-judge** scoring.
- Shipped the layer as a feature-flagged, opt-in capability across 3 runtime modes, with per-document fallback ensuring backward compatibility with production behavior.

Software Engineer Intern, *Trading Technologies Pvt. Ltd.* *Dec 2024 – Jun 2025*
- Automated more than **80%** of manual regression test cases for the SCORE compliance product, reducing execution time from **6 hours to under 1 hour**.
- Developed Python scripts to validate critical financial data transformations (KRM to RISK, WFE pipelines) and automated **RESTful API** validation, reducing manual verification effort by **70%**
- Built scalable automation frameworks integrated with **AWS** (S3, DynamoDB, EC2, EMR, Glue Jobs), accelerating file ingestion and validation workflows by **35%**
- Collaborated cross-functionally with QA and engineering teams to standardize test automation practices, reducing release-cycle bottlenecks by **40%**

Projects

Nanoserve: Single-GPU LLM Inference Engine — Python, PyTorch, CUDA, FlashInfer, Hugging Face, vLLM
- Built an **end-to-end Qwen2.5 inference engine** with custom model execution and SafeTensors loading, **physical paged KV caching, continuous batching, recompute preemption, and an OpenAI-style HTTP/SSE streaming API.**
- Integrated an opt-in FlashInfer paged-decode backend that increased measured in-window throughput by **81% (75.4 → 136.6 output tokens/sec), reduced p99 time-to-first-token from 19.2 s to 145 ms**, and **eliminated 92/387 request failures** on a pinned Qwen2.5-1.5B workload running on an NVIDIA L40S.
- Built a repeatable benchmarking system to measure inference speed, reliability and resource usage; checked model execution against Hugging Face and **replayed 3,691 requests against a same-session vLLM baseline with zero missing metrics or client scheduling violations.**

PolicyBot: Full-Stack HR RAG Assistant
- Shipped a full-stack **RAG** assistant (**FastAPI**, **React**) live on Render/Vercel with one **JWT** role-based access and audit layer shared by the REST API, chat agent and a **Slack** bot
- Built hybrid **BM25** + dense retrieval with cross-encoder reranking (**Recall@1 92.5%**, **MRR 0.958** on 67 labeled questions); traced the last retrieval miss to a chunking bug that stripped section headers, fixed it, and added a groundedness check that blocks unsupported numbers in answers
- Built interchangeable hand-rolled **ReAct** and **LangGraph** agent engines, the latter with per-user conversation memory; fixed a production bug in both that wrote duplicate leave requests by making successful tool calls terminal.

Fitbit-MCP: Personal Health Data Platform
- Built a local-first **MCP** server (13 tools) that syncs Fitbit / Google Health data into **SQLite**, powering a terminal dashboard, phone app and LLM-written daily coaching notes that I use every day
- Validated my recovery score against **WHOOP**'s (Pearson r = 0.97); diagnosed a constant offset with **Bland–Altman** analysis and recalibrated, cutting mean absolute error from 15.6 to 6.9 points and raising band agreement from 46% to 85%

Skills

Languages: Python, C/C++, Java, JavaScript, SQL
AI/ML: PyTorch, Hugging Face, Google ADK, LangChain, LangGraph, RAG, MCP, BM25, Cross-Encoder Reranking
LLM Inference: FlashInfer, vLLM, KV Caching, Continuous Batching, Performance Benchmarking
Web & APIs: FastAPI, Pydantic, Node.js, React, Tailwind CSS, REST APIs, JWT, Role-Based Access Control
Model Training: Supervised Fine-Tuning (SFT), LoRA, QLoRA, Parameter-Efficient Fine-Tuning (PEFT)
Cloud & Data: AWS (S3, SQS/SNS, EC2, DynamoDB, EMR, Glue), Docker, CI/CD, Redis, MongoDB, MySQL